

ME602T Assignment 4

Algorithm

Step 1: Declare and initialise the required variables to store length of domain, number of elements in X and Y directions and delta X and delta Y.

Step 2: Declare a matrix of dimensions $(m + 1) \times (n + 1)$ to store the temperature values at each node.

Step 3: Iterate over the entire matrix using nested for loops and assign the temperature values according to the following conditions

```
if i<50 && j<50 t_ij = 1
if i>=50 && j>= 50 t_ij = 0.5
else t_ij = 0.25
```

Step 4: Open an output filestream to write grid to .dat file named `grid.dat`

Step 5: Iterate over the entire grid using nested for loops. The X coordinate is determined by `dx*i` and Y coordinate by `dy*j`.

Step 6: The calculated coordinates are written to the output stream with a newline character after every vertex.

Step 7: Close the filestream and print confirmation to console.

Step 8: Open another output file stream to write temperatures to .dat file, named `T.dat`

Step 9: Iterate over the entire grid using nested for loops and calculate the X and Y coordinates of the vertex.

Step 10: The calculated coordinates along with the temperature at that vertex is written to the .dat file with a newline character after every vertex.

Step 11: Close the filestream and print confirmation to console.

Step 12: The .dat files output from the program are brought into MATLAB to plot the grid and temperature colour map.

Program Output

Input:

`.\assignment_4.exe`

Output:

Grid written successfully.

T written to file successfully.

Grid and Temperature colourmap plots

